

Cluster Robust Variance Estimation

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Introduction

- ▶ The variance of the sum of 2 random variables X_1 and X_2 is:

$$\text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot \underbrace{\text{cov}(X_1, X_2)}_{=0 \text{ if } X_1 \perp X_2}$$

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- ▶ The OLS estimator is the true parameter plus the average across individuals of the individual score

$$\hat{\beta}_{\text{ols}} = \beta + \frac{\sum_i X_i u_i}{\sum_i X_i^2}$$

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Which is the problem?

- ▶ Inference with correlation among cluster of individuals
- ▶ Usual White Variance estimator is not consistent

Literature review

Why Cluster?

- ▶ Traditional view: adjust for **within-cluster error correlation**.
- ▶ Bias in OLS S.E. grows with: strong error correlation, correlated regressors, large clusters.
- ▶ CRVE relies on **large G** asymptotics.

Fixed G

- ▶ Ibragimov and Müller (2010): inference can use **cluster-level estimates**.
- ▶ If they are (approximately) independent & Gaussian a **t-test** with t_{G-1} is valid even with heterogeneous variances.

The Design-Based Argument

- ▶ Clustering matters because of **sampling or assignment design**, not residual correlation.
- ▶ Residual correlation is **neither necessary nor sufficient**.
- ▶ If neither sampling nor treatment assignment is clustered **do not cluster**.
- ▶ CRVE is often conservative unless effects are homogeneous or clusters are rarely observed.

Heterogeneous effect

$$y_{ig} = \beta_g \mathbf{X}_{ig} + \gamma \mathbf{Z}_g + u_{ig}$$

$$\beta_g = \begin{bmatrix} \alpha_g \\ \beta_g \end{bmatrix}$$

Estimand

- ▶ In which parameter are we interested?



β_{APE}

Average Partial Effect

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Ibragimov, Rustam, and Ulrich K Müller. 2010. "T-Statistic Based Correlation and Heterogeneity Robust Inference." *Journal of Business & Economic Statistics* 28 (4): 453–68.